

Whole Human Chromosome TMRCA Decoding via Curriculum Learning with Mamba

fastext

Abstract

We describe a curriculum learning strategy to train a Mamba-based encoder–decoder model for decoding time to most recent common ancestor (TMRCA) across entire human chromosomes at 2 kb resolution. The key challenge is that the largest chromosome (chr1, 249 Mb) requires 124,500 sequence windows—far beyond what can be efficiently simulated in bulk. Our approach progressively increases sequence length across four training stages (50→100→150→250 Mb), using many replicates at shorter lengths and fewer at longer lengths. The Mamba architecture’s linear scaling with sequence length, combined with learnable positional embeddings that can be extended between stages, makes this curriculum natural and efficient.

1 The Challenge: Chromosome-Scale Inference

Human chromosomes range from 47 Mb (chr21) to 249 Mb (chr1). At 2 kb window resolution, this corresponds to 23,500–124,500 windows per chromosome. Coalescent simulation at these lengths is computationally expensive, and generating hundreds of training replicates at full chromosome scale is infeasible.

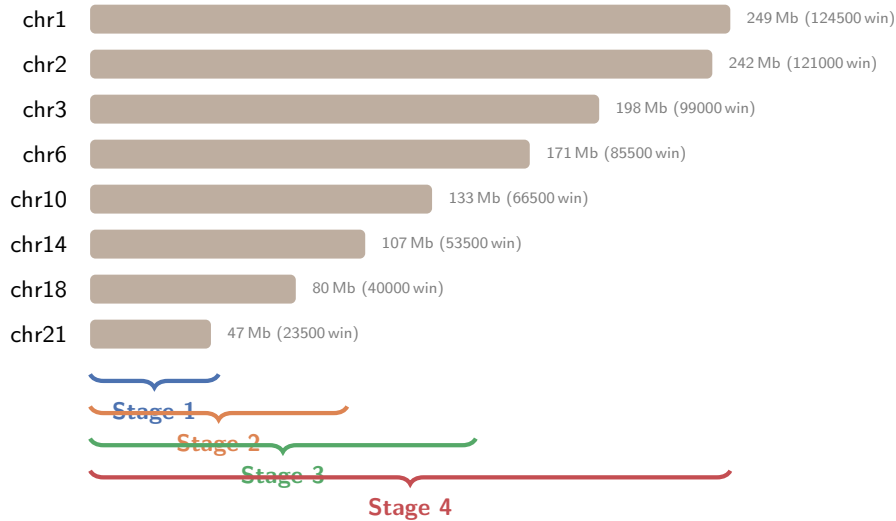


Figure 1: Human chromosome lengths and the sequence lengths covered by each curriculum stage. Stage 4 (250 Mb) covers the full extent of chr1.

2 Architecture: Mamba Encoder–Decoder

The model processes site frequency spectrum (SFS) features computed in non-overlapping 2 kb windows along the genome. A multi-scale convolutional stem encodes each window independently, then a stack of Mamba layers captures long-range dependencies along the chromosome.

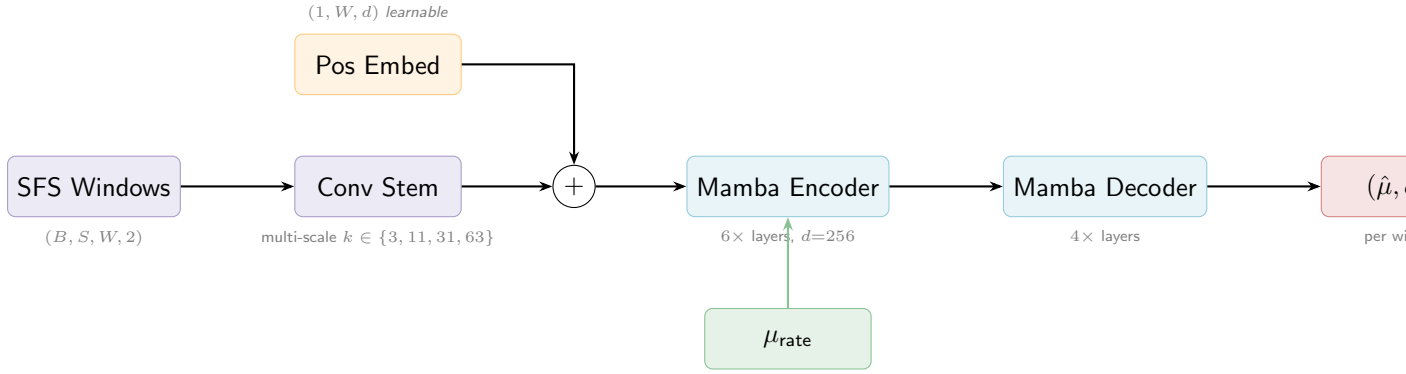


Figure 2: Model architecture. SFS features from each 2kb window are encoded by a multi-scale convolutional stem, combined with learnable positional embeddings, then processed by a Mamba encoder–decoder stack. The model outputs a Gaussian $(\mu, \log \sigma^2)$ per window for TMRCA prediction.

Key property: linear scaling. Unlike Transformers whose attention is $O(W^2)$, Mamba’s selective state space layers are $O(W)$ in sequence length W . This makes chromosome-scale sequences (up to 125k windows) tractable with constant per-window cost.

3 Curriculum Learning Strategy

We train in four stages of increasing sequence length. Each stage fine-tunes from the previous checkpoint, with the number of training replicates decreasing as simulation cost increases.

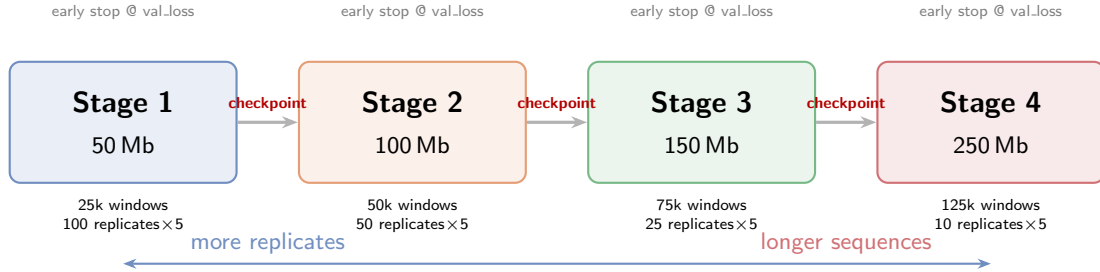


Figure 3: Curriculum learning pipeline. Each stage trains until early stopping triggers (patience = 20 on validation loss), then the checkpoint is used to initialize the next stage with extended positional embeddings. The trade-off between replicates and sequence length reflects simulation cost.

Table 1: Curriculum stage configuration. All stages use sample sizes $n \in \{10, 25, 50, 100, 200\}$ (“ $\times 5$ ”), 2 kb windows, batch size 4, gradient accumulation 64.

Stage	Seq. Length	W (windows)	Replicates/size	Preset	Sim Partition	Train Partition
1	50 Mb	25,000	100	base_25k	genoa (2d)	dgx-b200 (7d)
2	100 Mb	50,000	50	base_50k	genoa (2d)	dgx-b200 (7d)
3	150 Mb	75,000	25	base_75k	genoa (4d)	dgx-b200 (7d)
4	250 Mb	125,000	10	base_125k	genoa (4d)	dgx-b200 (7d)

4 Positional Embedding Extension

The positional embedding is a learnable parameter of shape $(1, W, d)$. At each stage transition, we must extend it to accommodate the larger sequence length. Our approach preserves existing positions and initializes new ones:

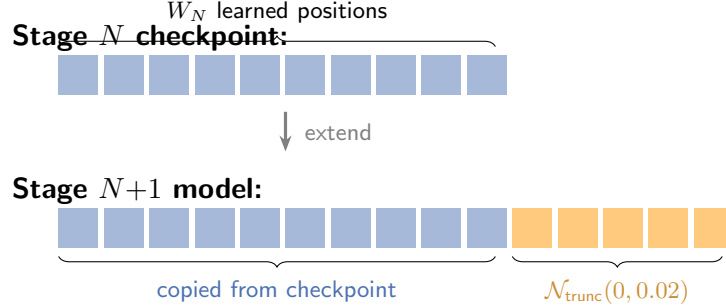


Figure 4: Positional embedding extension. Existing positions (blue) are preserved exactly from the checkpoint. New positions (orange) are initialized from a truncated normal distribution, matching the original initialization scheme. No interpolation is needed since positions represent fixed 2 kb genomic intervals.

This is more principled than interpolation for our setting: each position corresponds to an absolute genomic coordinate at fixed 2 kb spacing. The model has already learned meaningful representations for positions 1– W_N ; we simply add new positions W_N+1 to W_{N+1} with fresh initialization.

5 Simulation Design

Training data is generated using `stdpopsim` with the `HomSap` species. This provides a natural mixture of demographic histories:

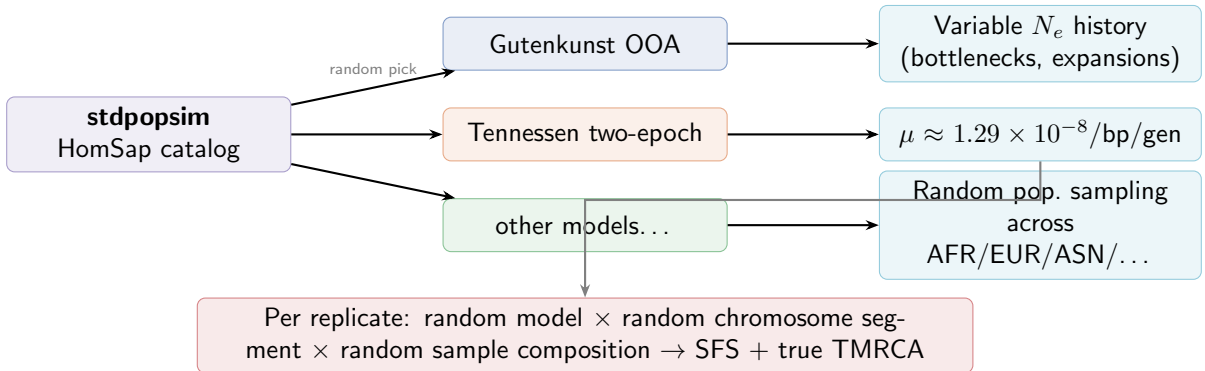


Figure 5: Simulation design. Each replicate randomly draws a demographic model from the `stdpopsim` `HomSap` catalog, a random autosomal segment of the target length, and a random allocation of samples across populations. This produces diverse training data reflecting real human genetic variation.

6 Training Configuration

Training hyperparameters.

- Batch size: 4 (actual) $\times$ 64 (gradient accumulation) = 256 effective

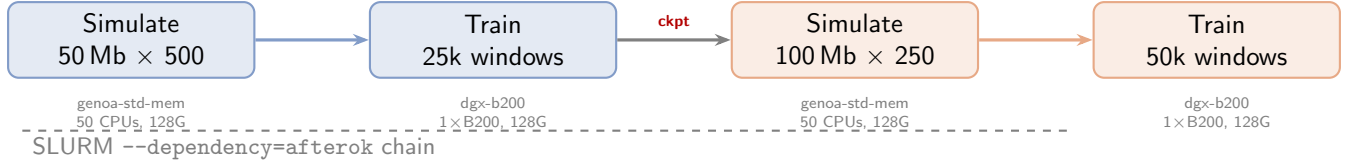


Figure 6: SLURM execution pipeline. Simulation and training jobs alternate, connected by dependency chains. Each training job produces a checkpoint consumed by the next stage’s training job.

- Learning rate: 3×10^{-4} with cosine decay
- Loss: Beta-NLL ($\beta = 0.5$) for heteroscedastic TMRCA prediction
- Early stopping: patience 20 epochs on validation loss ($\delta_{\min} = 10^{-4}$)
- Precision: bf16-mixed
- Max epochs: 200 (early stopping typically triggers well before)

7 Expected Outcomes

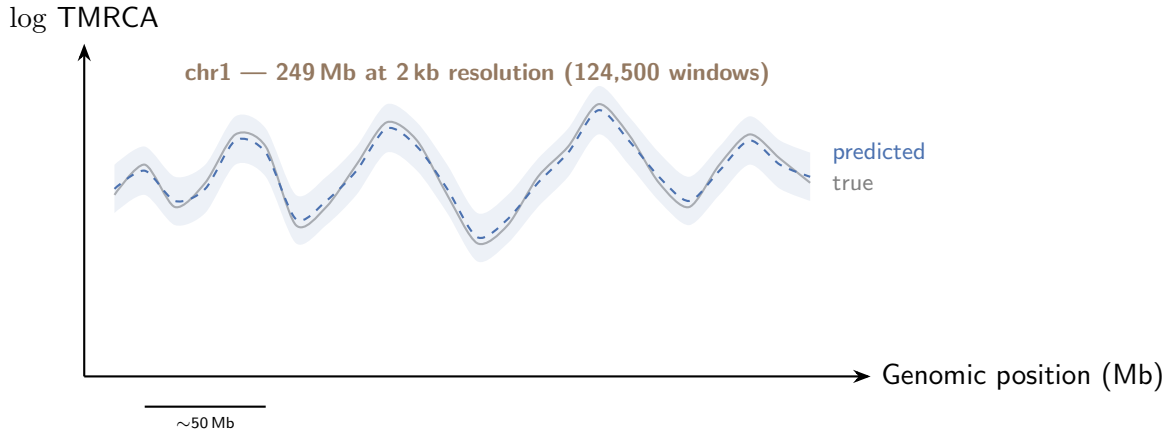


Figure 7: Schematic of expected output: per-window TMRCA prediction with calibrated uncertainty ($\hat{\mu} \pm 1.96\hat{\sigma}$) across an entire human chromosome. The model produces both point estimates and confidence intervals, enabling downstream inference of selection, introgression, and demographic history.

Applications. Chromosome-scale TMRCA maps at 2 kb resolution enable:

1. Detection of selective sweeps as local TMRCA troughs
2. Identification of archaic introgression via elevated TMRCA
3. Fine-scale recombination rate inference from TMRCA transitions
4. Demographic inference from the genome-wide TMRCA distribution

The curriculum learning approach makes this feasible without requiring chromosome-scale simulation at full volume, leveraging the Mamba architecture’s ability to generalize learned local patterns to longer contexts.